SETS

- **Set**: Set is a collection of well defined distinct objects. It is denoted by capital letters of English alphabets as A, B, C, D,… etc. The members of the set are called as Elements. The elements of the set are denoted by small letters of English alphabets. If an element a belongs to the set A, then it is denoted as $a \in A$ and if an element a does not belongs to the set A, then it is denoted as $a \notin A$.

- **Laws of Algebra of a Set**:

(1) **Idempotent Laws**: For any set A,
- (a) $A \cup A = A$.
- (b) $A \cap A = A$.

(2) **Identity Laws**: For any se A,
- (a) $A \cup$ ¤ $= A$. Here ¤ is an empty set.
- (b) $A \cap U = A$. Here U is an universal set.

(3) **Communicative Laws**: For any two sets A and B,
- (a) $A \cup B = B \cup A$.
- (b) $A \cap B = B \cap A$.

(4) **Associative Laws**: For any three sets A, B and C,
- (a) $(A \cup B) \cup C = A \cup (B \cup C)$.
- (b) $(A \cap B) \cap C = A \cap (B \cap C)$.

(5) **Distributive Laws**: For any three sets A, B and C,
- (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
- (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(6) **De-Morgan's Laws**: For any two sets A and B,
- (a) $(A \cup B)' = A' \cap B'$.
- (b) $(A \cap B)' = A' \cup B'$.

(7) **Results on operation on sets**: For any sets A, B and C,
- (a) $A - B = A \cap B'$.
- (b) $B - A = B - A'$.
- (c) $A - B = A \Leftrightarrow A \cap B =$ ¤.
- (d) $(A - B) \cup B = A \cup B$.
- (e) $(A - B) \cap B =$ ¤.
- (f) $A \subseteq B \Rightarrow B' \subseteq A'$.
- (g) $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.
- (h) $A - (B \cup C) = (A - B) \cap (A - C)$.
- (i) $A - (B \cap C) = (A - B) \cup (A - C)$.
- (j) $A \cap (B - C) = (A \cap B) - (A \cap C)$.
- (k) $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$.

(8) **Results on Number of Elements in Sets**: If A, B and C are finite sets and U is the finite Universal set, then:

(l) $n(A\cup B)=n(A)+n(B)-n(A\cap B)$.

(a) If $n(A\cup B)=n(A)+n(B)\Leftrightarrow$ A and B are disjoint sets.

(b) $n(A-B)=n(A)-n(A\cap B)$.

(c) $n(A-B)+n(A\cap B)=n(A)$.

(d) $n(A\Delta B)=n(A)+n(B)-2n(A\cap B)$.

(e) $n(A\cup B\cup C)=n(A)+n(B)+n(C)-n(A\cap B)-n(B\cap C)-n(C\cap A)+n(A\cap B\cap C)$.

(f) No. of elements in exactly two of the sets A, B and C =

$$n(A\cap B)+n(B\cap C)+n(C\cap A)-3n(A\cap B\cap C).$$

(g) No. of Elements in exactly one of the sets A, B and C =

$$n(A)+n(B)+n(C)-2n(A\cap B)-2n(B\cap C)-2n(C\cap A)+3n(A\cap B\cap C).$$

(h) $n(A'\cup B')=n\left((A\cap B)'\right)=n(U)-n(A\cap B)$.

(i) $n(A'\cap B')=n\left((A\cup B)'\right)=n(U)-n(A\cup B)$.

(9) **Cartesian Product of Two Sets**: Let A and B are two sets, then their Cartesian product is defined as $A\times B=\{(x,y):x\in A, y\in B\}$. Similarly Cartesian product of $n\geq 2$ sets $A_1, A_2, A_3, ..., A_n$ is defined by

$$A_1\times A_2\times A_3\times ...\times A_n=\prod_{k=1}^{n} A_k=\{(a_1,a_2,...,a_n):a_1\in A_1, a_2\in A_2, a_3\in A_3,...,a_n\in A_n\}.$$

(10) **Properties of Cartesian Product of Sets**:

(a) $A\times B\neq B\times A$.

(b) $A\times ¤ = ¤\times A = ¤$.

(c) If A and B are finite sets, then $n(A\times B)=n(A)\times n(B)$.

(d) If $A\subseteq B$, then $A\times C\subseteq B\times C$.

(e) $A\times(B\cup C)=(A\times B)\cup(A\times C)$.

(f) $A\times(B\cap C)=(A\times B)\cap(A\times C)$.

(g) $A\times(B-C)=(A\times B)-(A\times C)$.

(h) $A\subseteq B, C\subseteq D\Rightarrow A\times C\subseteq B\times D$.

(i) $(A\times B)\cup(C\times D)\subseteq(A\cup C)\times(B\cup D)$.

(j) $(A\times B)\cap(C\times D)\subseteq(A\cap C)\times(B\cap D)$.

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